HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79SP

STOCHASTIC PROCESSES

IN-PERSON EXAMINATION

December Examination and Assessment Diet 2024

Duration: 2 hours

Total marks available: 80

Answer all FIVE questions

Attached: Formulae sheet included
Supplied: No additional material supplied
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

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Some Absolutely Continuous Distributions

Name	Notation	p.d.f.	range	m.g.f.	mean	variance
Uniform	$U(a,b)$	$\frac{1}{b-a}$	$a \le x \le b$	$\frac{e^{b\theta}-e^{a\theta}}{(b-a)\theta}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu,\sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$	$-\infty < x < \infty$	$e^{\mu\theta+\frac{1}{2}\sigma^2\theta^2}$	μ	σ^2
Gamma	$\Gamma(t,\lambda)$	$\frac{\lambda^t}{\Gamma(t)}x^{t-1}e^{-\lambda x}$	$0<x<\infty$, $t,\lambda>0$	$\left(\frac{\lambda}{\lambda-\theta}\right)^t$	$\frac{t}{\lambda}$	$\frac{t}{\lambda^2}$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$0<x<\infty$, $\lambda>0$	$\frac{\lambda}{\lambda-\theta}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Chi-square	χ^2_n	$\frac{1}{2^{n/2}\Gamma(n/2)}x^{n/2-1}e^{-x/2}$	$0<x<\infty$, $n>0$ integer	$(1-2\theta)^{-n/2}$	n	$2n$
Beta	$\beta(\alpha_1,\alpha_2)$	$\frac{\Gamma(\alpha_1+\alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)}x^{\alpha_1-1}(1-x)^{\alpha_2-1}$	$0<x<1$, $\alpha_1,\alpha_2>0$	not useful	$\frac{\alpha_1}{\alpha_1+\alpha_2}$	$\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2(\alpha_1+\alpha_2+1)}$
Cauchy	–	$\frac{\lambda}{\pi(\lambda^2+(x-\mu)^2)}$	$-\infty<x<\infty$, $\lambda>0$	doesn't exist but char. fn. is $e^{i\mu\theta-\lambda\lvert\theta\rvert}$	no moments exist	
Multivariate Normal	$N(\mu,\Sigma)$	$\frac{\exp\left\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right\}}{(2\pi)^{d/2}\det(\Sigma)^{1/2}}$	$x\in\mathbb{R}^d$	$e^{\mu^T\theta+\frac{1}{2}\theta^T\Sigma\theta}$	μ	covariance matrix Σ

Some Discrete Distributions

(All distributions below have integer-valued ranges)

Name	Notation	prob. fn.	range	m.g.f.	mean	variance
Binomial	$\text{Bin}(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$0\le x\le n$ $0<p<1$	$[(1-p)+pe^\theta]^n$	np	$np(1-p)$
Poisson	$\text{Pois}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	$0\le x<\infty$ $\lambda>0$ real	$e^{\lambda(e^\theta-1)}$	λ	λ
Hyper-geometric	$\text{Hgeo}(n,a,b)$	$\frac{\binom{a}{x}\binom{b}{n-x}}{\binom{a+b}{n}}$	$0\le x\le n$ $a,b>0$ integers	not useful	$\frac{na}{a+b}$	$\frac{nab(a+b-n)}{(a+b)^2(a+b-1)}$
Negative binomial	$\text{Nbin}(k,p)$	$\binom{x-1}{k-1}p^k(1-p)^{x-k}$	$k\le x<\infty$, $0<p<1$	$p^k[e^{-\theta}-(1-p)]^{-k}$	k/p	$k(1-p)/p^2$
Geometric	$\text{Geo}(p)$	$p(1-p)^{x-1}$	$1\le x<\infty$ $0<p<1$	$p[e^{-\theta}-(1-p)]^{-1}$	$1/p$	$(1-p)/p^2$

Multinomial: multivariate extension of binomial

Prob. fn. is given by

$$f(x_1,x_2,\ldots,x_k)=\frac{n!}{x_!x_2!\ldots x_k!x_{k+1}!}p_1^{x_1}p_2^{x_2}\ldots p_k^{x_k}p_{k+1}^{x_{k+1}},\quad 0\le x_i\le n,\ \ 0<p_i<1\ \ \forall i=1,2,\ldots,n+1$$

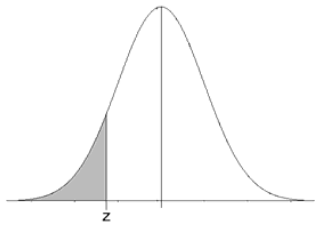
where $p_1+p_2+\cdots+p_{k+1}=1$ and $x_1+x_2+\cdots+x_{k+1}=n$.

The m.g.f. is given by

$$[p_1e^{\theta_1}+p_2e^{\theta_2}+\cdots+p_ke^{\theta_k}+p_{k+1}]^n.$$

Moments: $E(X_i)=np_i$, $\text{Var}(X_i)=np_i(1-p_i)$, $\text{cov}(X_i,X_j)=-np_ip_j$, $i,j=1,2,\ldots,n+1$.

Standard Normal Cumulative Probability Table



Cumulative probabilities for NEGATIVE z-values are shown in the following table:

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

1. (a) Let $S_n = a + \sum_{i=1}^{n} \xi_i$ be the simple random walk with initial state $S_0 = a$, where a is a positive integer, and the random variables ξ_i are independent and identically distributed with common distribution

$$\mathbf{P}(\xi_i = 1) = p > 0, \quad \mathbf{P}(\xi_i = -1) = q > 0, \quad p + q = 1.$$

Assume in addition that $q > p$. Let

$$T_a = \min\{n \geq 1 \ : \ S_n \leq 0\}$$

be the time to "ruin". Prove the mean time to "ruin", $\mathbf{E}T_a$, is equal to

$$\mathbf{E}T_a = \frac{a}{1-2p}.$$

[6 marks]

(b) Consider a random walk $S_n = a + \sum_{i=1}^{n} \xi_i$ with initial state a, but now assume there is the possibility of jumps of zero size,

$$\mathbf{P}(\xi_i = 1) = p > 0, \quad \mathbf{P}(\xi_i = -1) = q > 0, \quad \mathbf{P}(\xi_i = 0) = r > 0, \quad p + r + q = 1.$$

Prove the mean time to "ruin" in this case, $\mathbf{E}T_a$, is equal to

$$\mathbf{E}T_a = \frac{a}{q-p}.$$

[4 marks]

(c) An insurance company has initial capital 5. Each year it receives premium income of 1 and pays claims. The total claim paid in year i is Y_i where $\{Y_i\}$ is a sequence of independent identically distributed random variables.

i. Assume that each of the Y_i's has the following distribution

$$\mathbf{P}(Y_i = 0) = 1/20, \quad \mathbf{P}(Y_i = 1) = 7/20, \quad \mathbf{P}(Y_i = 2) = 12/20.$$

Find the expected time to ruin in this case.

[3 marks]

ii. Assume now that the common distributions of the Y_i's is

$$\mathbf{P}(Y_i = 0) = 2/20, \quad \mathbf{P}(Y_i = 1) = 7/20, \quad \mathbf{P}(Y_i = 2) = 5/20, \quad \mathbf{P}(Y_i = 3) = 6/20.$$

Provide either an upper or a lower estimate on the expected ruin time in this case, based on comparison with another random walk; with an explanation of your reasoning.

[3 marks]

[Total: 16 marks]

2. (a) Prove that if a time-homogeneous Markov chain has a finite state space and is irreducible and aperiodic, then it has a unique stationary distribution which coincides with the limiting distribution, for any initial state.
Hint: You are allowed to refer to the following result: For an irreducible and aperiodic Markov chain with finite or countable state space $\mathcal{S}$ and transition matrix $P = \{p_{i,j}\}$, the following holds: for any $i, j \in \mathcal{S}$, the limit $\omega_j = \lim_{n\to\infty} p_{i,j}^{(n)}$ exists and does not depend on i. Here $p_{i,j}^{(n)}$ are n-step transition probabilities.

[7 marks]

(b) A motorist purchases an insurance policy each year. There are four levels of discount: level 0 (0% discount), level 1 (15% discount), level 2 (30% discount), or level 3 (50% discount). In the following year he moves one level up (or remains at level 3 if already there) provided there has been no claim during the year. Otherwise he moves one level down (or remains at level 0 if already there) if there was only one claim during the year, and moves two levels down (or moves to level 0 if he is either at level 0 or at level 1) if there were two or more claims.

Assume that, independently of everything else, within each year the motorist makes one claim with probability 0.1 and two or more claims with probability 0.01.

i. Find the transition matrix for the Markov chain that models this policy.

[2 marks]

ii. Prove that the Markov chain is irreducible and aperiodic.

[2 marks]

iii. Apply the result from item (2a) above to find the stationary distribution and a limiting distribution if $X_0 = 0$.

[4 marks]

iv. Assume now that the insurance company decides to modify the policy in the following way: if the motorist gets to level 0, he stays there forever; similarly, if he gets to level 3, he stays there forever. All other policy details remain the same.

A. Find the transition matrix of the new Markov chain, classify all its states, and find all associated stationary distributions.

[4 marks]

B. Find absorption probabilities for each closed class and each initial state.

[4 marks]

[Total: 23 marks]

3. Consider a Galton-Watson branching process $\{X_n, n \geq 0\}$ with initial state $X_0 = k \geq 1$. Let p_k denote the probability that an individual has k offspring and suppose that the offspring distribution is given by

$$p_0 = \frac{3(1-p)}{4}, \quad p_1 = \frac{1+2p}{4}, \quad p_2 = \frac{p}{4}, \quad \text{and} \quad p_l = 0, \quad \text{for all other} \quad l.$$

Let $\varepsilon(k, p)$ be the extinction probability of this process.

(a) Find all values of parameter p, for which $\varepsilon(k, p)$ is equal to 1.

[3 marks]

(b) Find all values of parameter p, for which $\varepsilon(k, p)$ is equal to 0.

[2 marks]

(c) Find the extinction probability if

i. $p = 5/6$ and $k = 1$;

[4 marks]

ii. $p = 5/6$ and $k = 5$.

[2 marks]

(d) Let $p = 0$ and let $T_k = \min\{n \geq 0 \ : \ X_n = 0 \mid X_0 = k\}$.
Find $\mathbf{P}(T_1 > 3)$ and $\mathbf{E}T_1$.

[3 marks]

[Total: 14 marks]

4. An insurance company receives claims according to a Poisson process with rate $\lambda = 5$ per day. Assume that the claim sizes are i.i.d. and take the values $20, 70$, and 120 with corresponding probabilities $1/4$, $1/4$, and $1/2$.

(a) Let X be the number of claims of size 120 that are received within four consecutive days. What is the distribution of X ?

[1 mark]

(b) Consider again four consecutive days. What is the conditional probability that during the first day there were 3 claims of size 20 given that there were 5 claims in total within four days ?

[5 marks]

(c) Determine the probability that the fifth claim of size 120 arrives before the second claim of size 20.

[4 marks]

(d) Derive the formula for the probability that, within two days, the total of all claims is less than 80 (no need to give the final arithmetic answer).

[5 marks]

[Total: 15 marks]

5. (a) Let a continuous-time Markov process $X(t)$ have a transition rates matrix Q of the form

$$Q = \begin{pmatrix} -4\lambda & 3\lambda & \lambda \\ \lambda & -2\lambda & \lambda \\ \lambda & \lambda & -2\lambda \end{pmatrix}$$

where λ is a positive number.

i. For this Markov process, find all stationary distributions and limiting distributions for all initial values. Justify your answers.

[4 marks]

ii. Consider the embedded Markov chain $X_n, n \geq 0$. Find all its stationary distributions and limiting distributions for all initial values. Justify your answers.

[4 marks]

iii. Do the answers in the two previous questions depend on λ? Provide your comment.

[1 mark]

(b) Consider instead a continuous-time Markov process $Y(t)$ with transition rate matrix

$$Q = \begin{pmatrix} -4\lambda & 3\lambda & \lambda \\ \lambda & -4\lambda & 3\lambda \\ 3\lambda & \lambda & -4\lambda \end{pmatrix}$$

Find all stationary and all limiting distributions for this Markov process and for its embedded Markov chain without any calculations. Justify your reasons.

[3 marks]

[Total: 12 marks]

END OF PAPER